INCH-POUND
MIL-DTL-16377/71A(SH)
26 May 2023
SUPERSEDING
MIL-F-16377/71(SH)
14 May 1982

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT AND SOLID-STATE EQUIVALENT, DETAIL LIGHTING FOR MIRRORS FOR SMALL BOATS, 8 WATTS, 120 VOLTS, 60 HERTZ, TOTALLY ENCLOSED, SYMBOLS 354 AND 354SSL

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

Unless otherwise noted, all dimensions are in inches. For tolerances other than those specified herein, see 3.5.2 of MIL-DTL-16377.

Figures 1 through 4 provide information for the light fixtures covered by this specification sheet. See the requirements sections for specific references to these figures.

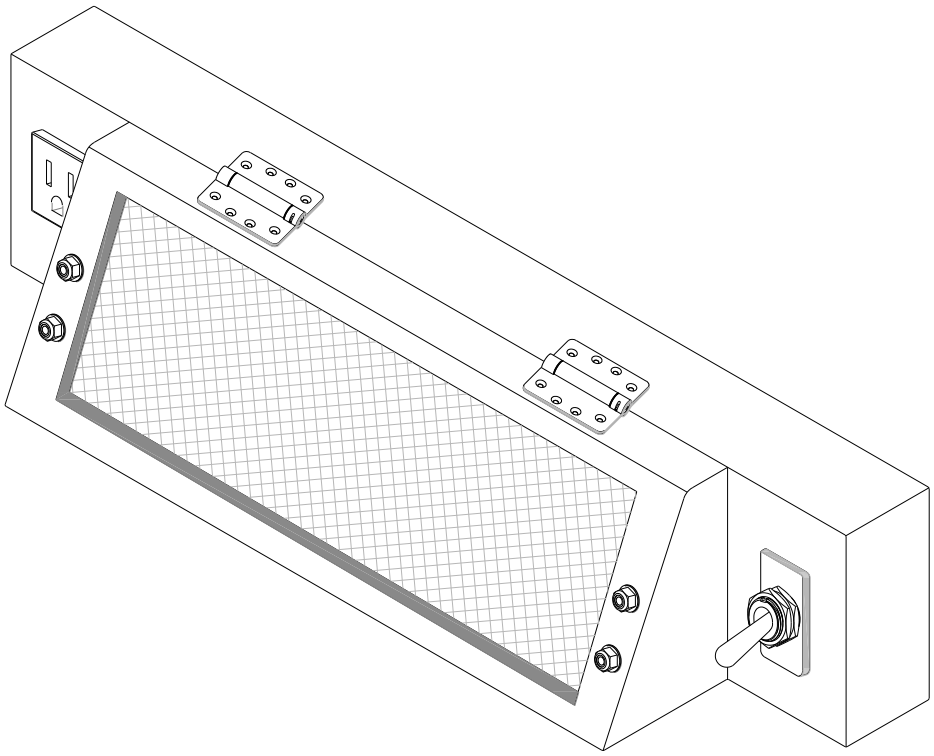


FIGURE 1. Typical view of symbol 354.

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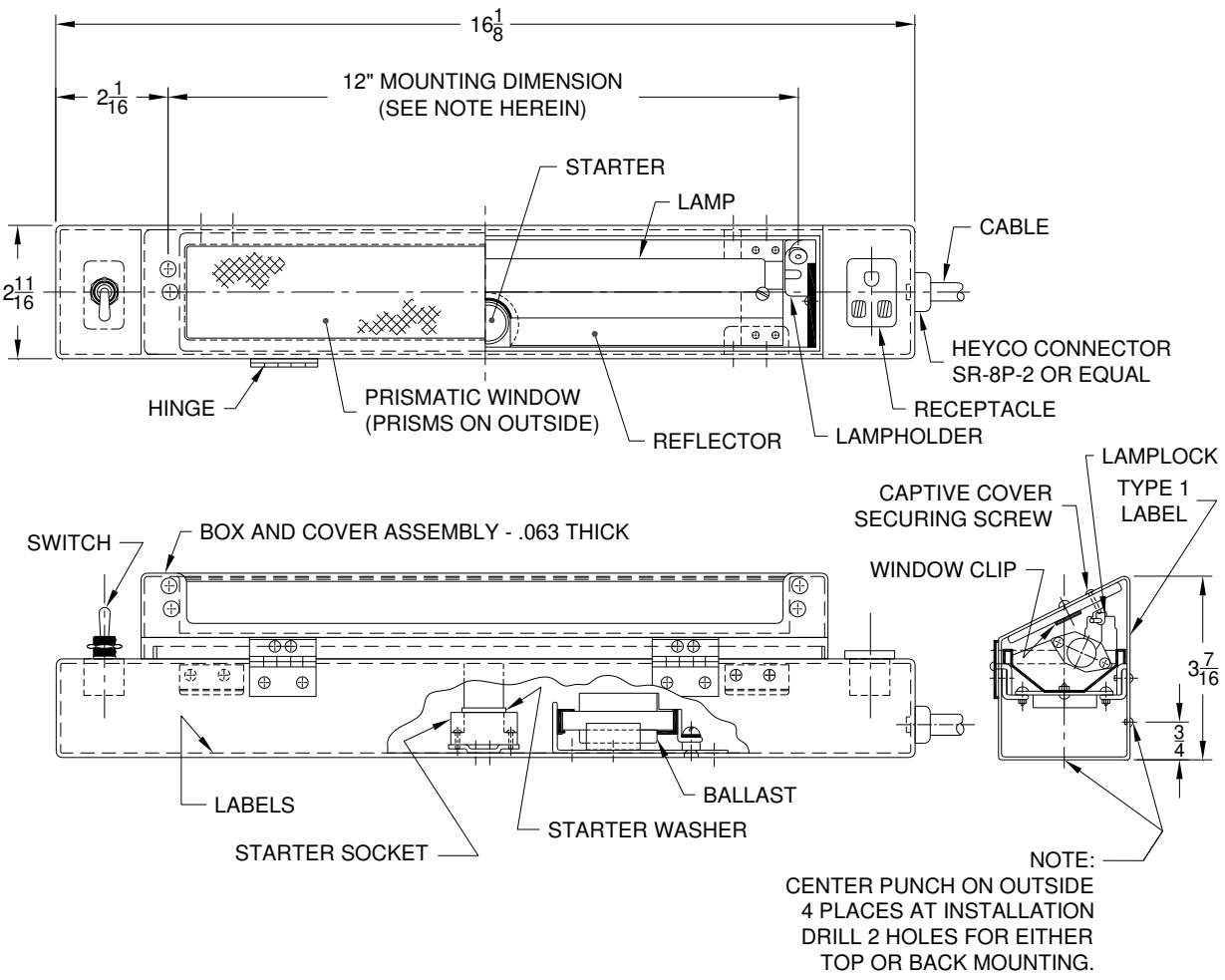
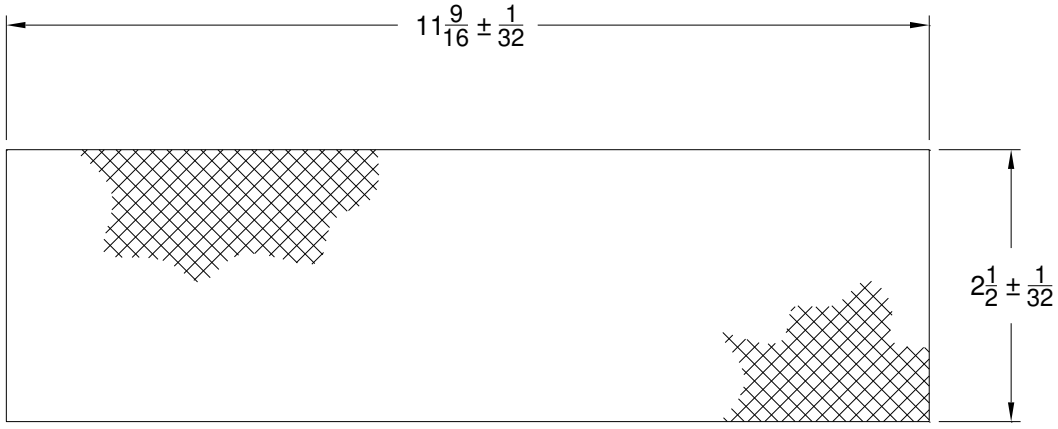


FIGURE 2. Typical section views of symbol 354.



MATERIAL FOR WINDOW SHALL BE 1/8 INCH THICK, CLEAR, PRISMATIC (ONE SIDE ONLY), HEAT RESISTANT, CAST ACRYLIC PLASTIC SHEET, TYPE CLEAR-2 OF MIL-DTL-24191.

FIGURE 3. Prismatic window.

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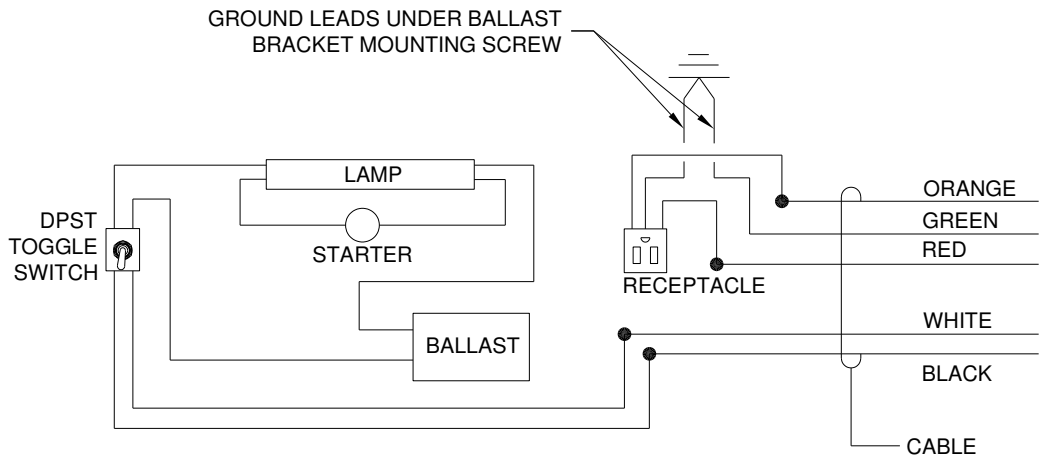


FIGURE 4. Wiring diagram.

REQUIREMENTS

CLASSIFICATION

Type: I
Class: 1

SYMBOL NUMBER

354: Complete fixture, 16⅛ inches in length with one 8-watt, cool white fluorescent lamp, clear prismatic window, and with toggle switch and single receptacle.

DESIGN

Electrical rating:	8 watts, 120 volts, 60 hertz.
Dimensions and configuration:	See figures 1 through 3.
Prismatic window:	See figure 3.
Wiring diagram:	See figure 4.
Photometrics:	See candlepower distribution curve for symbol 79.1 fixture in MIL-DTL-16377/14.
Cable:	10 feet of type LSMHOF-7 in accordance with MIL-DTL-24643/7.
Receptacle:	Receptacle shall be Leviton part number 1374-051 (ivory) or equal, except that the ground contact shall be nickel-plated, 0.0003-inch thick.
Weight:	3 pounds, 8 ounces.
Enclosure:	Totally enclosed.
Switch:	Switch shall be toggle, DPST, in accordance with MS16569-1.
Finish:	Complete fixtures to be finished as same as for type I, class 1 fixtures of MIL-DTL-16377 as applied as shown in figures 1 and 3.
Designation and marking:	The following labels shall be furnished in accordance with MIL-DTL-16377. (a) Two type I. (b) One type IV. The information to be shown on the type IV label shall be: "USE THE FOLLOWING REPAIR PARTS ONLY." List all standard parts shown in table I and window assemblies of table IV by name and stock number only.
Standard parts:	The standard parts shown in table I shall be used in the construction of the fixtures specified herein.

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SOLID-STATE LIGHTING (SSL) REQUIREMENTS

CLASSIFICATION

Type: III
Class: 1

SYMBOL NUMBER:

354SSL: Complete solid-state fixture, white, with toggle switch and single receptacle.

DESIGN:

Electrical rating:	8 watts maximum, 120 volts, 60 hertz.
Dimensions:	See figures 2 through 4.
Cable:	10 feet of type LSMHOF-3 in accordance with MIL-DTL-24643/7.
Receptacle:	Receptacle shall be Leviton part number 1374-051 (ivory) or equal, except that the ground contact shall be nickel-plated 0.0003-inch thick.
Photometric data:	See candlepower distribution curve for symbol 79.1 fixture in MIL-DTL-16377/14.
Enclosure:	Totally enclosed.
Chromaticity:	See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit:	See 3.17 of MIL-DTL-16377.
Wiring:	Wiring shall maintain same control scheme as legacy fluorescent fixture.
Weight:	3 pounds, 8 ounces.
Designation and marking:	One type I label shall be furnished in accordance with MIL-DTL-16377.

EXAMINATIONS AND TESTS

Examinations and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:	
Complete fixtures:	See table II for type I and table IV for type III.
Window assemblies:	See table III.
Conformance inspection:	
Comparison inspection:	Same as the first article inspection.
Conformance inspection:	See MIL-DTL-16377.

GENERAL INFORMATION

Assignment of military part numbers and general information is shown in table IV.

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TABLE I. Standard parts.

Name	Specification or Military Part Number	National Stock Number (NSN)	Remarks
Lamp	W-L-116/3	6240-00-299-5546	Industry number F8T5/CW
Lamplock	M16377/40-001	6250-00-399-7039	---
Lampholder	UL 496	6250-00-377-8590	Supplement SB, miniature bipin
Startersocket	UL 496	6250-00-295-2738	Supplement SB, medium bipin starter socket
Starter	8W, ANSI C78.180, type III	6250-00-299-5962	Commercial designation: FS5
Ballast	M16377/45-001	6250-00-295-2735	---
Switch, toggle, DPST	MS16569-1	5930-00-719-9659	---

TABLE II. First article and comparison inspection for complete fixtures.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Shock	3.5.12	4.8.7	Grade B
Vibration	3.5.13	4.8.8	---
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Noise	3.5.14	4.8.9	---
Grounding circuit	3.6.13.3	4.8.11	---
Magnetic permeability	3.4.8	4.8.18	---

TABLE III. First article and comparison inspection for prismatic windows.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377
Examination	3.12	4.5
Optical uniformity	3.7.4.1.1	4.8.4

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TABLE IV. First article and comparison inspection for type III luminaires.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Power interface	3.6.18	4.8.21	---
Electromagnetic interference	3.6.14	4.8.16	---
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.16	4.8.3	1/
Shock	3.5.12	4.8.7	Grade B
Vibration	3.5.13	4.8.8	1/
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.14	4.8.3	1/
Noise	3.5.14	4.8.9	1/
Photometrics	3.7.1	4.8.6	Candlepower
Brightness	3.7.5	4.8.6.4	---
Chromaticity	3.7.4	4.8.6.5	---
Correlated color temperature	3.7.4.1.1	4.8.6.6	---
Lumen maintenance	3.7.11	4.8.20	---
Fail-safe circuit design	3.17	4.8.24	---
Optical uniformity	3.7.7.1	4.8.4	---
Stress relief	3.8.5.3	4.8.5	---
Magnetic permeability	3.4.8	4.8.18	---
Grounding circuit	3.6.13.3	4.8.11	---

NOTE:

1/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

TABLE V. Part number and general information.

Military Part Number M16377/71	Item Name	National Stock Number (NSN)	Remarks
354	Complete fixture, symbol 354	---	Figure 2
354SSL	Complete fixture, symbol 354SSL	---	---
001	Prismatic window	6210-00-643-1217	Figure 3

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CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 6210-2023-018)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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